

Welcome to Math 222!

You have worked hard to be here;
you belong here. Congratulations on your
achievements and welcome to the course.
I'm excited you are here and I hope it
will be a great semester.

Agenda

1. Introduction (me)
2. Introduction (the course)
3. Introduction (you)
4. Integration by Parts

I. Introduction (me)

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The most important things for you to know about me:

2. Introduction (course)

Course Topics

There are four topics discussed this semester:

- A. Techniques of Integration
- B. Differential Equations
- C. Infinite Sequences and Series
- D. Vectors

Exams

- Midterm 1 : Tuesday, February 28, 7:30pm - 9:00pm
- Midterm 2 : Tuesday, April 11, 7:30pm - 9:00pm
- Final : Monday, May 8, 12:25pm - 2:25pm

Weekly Schedule

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
DO		Lecture	Discussion	Lecture	Discussion	Lecture
DUE	module quizzes		Midterms	Homework	Discussion Quiz	

Course Information

All important course information will be relayed through Canvas.

Your ebook and WebAssign are available through Canvas

3. Introduction (you)

4. Integration By Parts

Activity: Classify the following integrals as

A i) can be integrated directly

B ii) can be integrated after doing some algebra

C iii) can be integrated with u-substitution

D iii.) We need a new technique to integrate it

a.) $\int x^2 dx$ ¹ b.) $\int \frac{1}{x\sqrt{x^2-1}} dx$ ³ c.) $\int \sec x + \tan x dx$ ¹

d.) $\int \ln x dx$ ⁴ e.) $\int \sqrt[3]{x^3 + 3\cos x} (x^2 - \sin x) dx$ ³

f.) $\int x e^{2x} dx$ ⁴

g.) $\int \left(\frac{1 + 2xe^x}{x} \right) dx$ ²

h.) $\int x \cos^2 x dx$ ⁴

Formula Sheet

Integrals

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \text{ for } n \neq -1$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int \sin(x) dx = -\cos(x) + C$$

$$\int \cos(x) dx = \sin(x) + C$$

$$\int \tan(x) dx = -\ln |\cos(x)| + C$$

$$\int \cot(x) dx = \ln |\sin(x)| + C$$

$$\int \sec(x) dx = \ln |\sec(x) + \tan(x)| + C$$

$$\int \csc(x) dx = -\ln |\csc(x) + \cot(x)| + C$$

$$\int \sec(x) \tan(x) dx = \sec(x) + C$$

$$\int \sec^2(x) dx = \tan(x) + C$$

$$\int \csc(x) \cot(x) dx = -\csc(x) + C$$

$$\int \csc^2(x) dx = -\cot(x) + C$$

$$\int a^x dx = \frac{a^x}{\ln(a)} + C$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x) + C$$

Trigonometric Formulas

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$\cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}$$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}$$

$$\sec^2(x) - 1 = \tan^2(x)$$

$$\sin(x) \cos(y) = \frac{1}{2} (\sin(x-y) + \sin(x+y))$$

$$\sin(x) \sin(y) = \frac{1}{2} (\cos(x-y) - \cos(x+y))$$

$$\cos(x) \cos(y) = \frac{1}{2} (\cos(x-y) + \cos(x+y))$$

Integration by Parts is a technique of integration based on the product rule. *The reverse-product rule.*

recall, if $u = f(x)$ $v = g(x)$, and diff'able :

$$\frac{d}{dx}[f(x)g(x)] = u'v + uv'$$

Integrating both sides yields

$$\int \frac{d}{dx}[f(x)g(x)] dx = \int [f'(x)g(x) + f(x)g'(x)] dx$$

$$\text{then, } f(x)g(x) = \int f'(x)g(x) dx + \int f(x)g'(x) dx$$

Int. by pte. formula

$$\text{then, } \int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx$$

and

$$\int u dv = uv - \int v du$$

Example: $\int x e^x dx$

$$dv = e^x dx$$

$$u = x$$

$$\int x e^x = x e^x - \int e^x dx$$

$$= x e^x - e^x + C$$

Activity: $\int x \sin x dx$

$u =$

Integration by Parts: Guidelines for choosing u and dv

- dx is always part of du
- dv should be easy to integrate
- u and dv are chosen so that $\int v du$ is no more difficult to integrate than the original integral
- If the new integral is more complicated, try different choices for u and dv